

PROBABILITY DISTRIBUTION FUNCTIONS AND DATA

Random Variable

A random variable is a variable whose value is decided by the outcome of a random experiment. It helps us represent uncertain events mathematically.

A random variable assigns a numerical value to each outcome of a random experiment and the type depends on the set of values it can take.

For rolling a die, a random variable X can take values 1, 2, 3, 4, 5, 6.

A random variable is **discrete** if it has a finite or countable number of possible outcomes that can be listed. These values are often represented by whole numbers. Ex: No. of heads in 10 coin flips.

The discrete random variable is defined by **PMF (Probability Mass Function)**.

A random variable is **Continuous** if it has an uncountable number of possible outcomes within a given interval (a, b) , means that each and every single point in the interval can be taken up as the outcome values.

Thus, $X = \{x: x \text{ belongs to } (a, b)\}$

Example of a Continuous Random Variable

- The height of an adult male or female. ($145 < x < 170$)
- The speed of a vehicle on a highway ($80 < x < 120$)

The continuous random variable is defined by **PDF (Probability Density Function)**

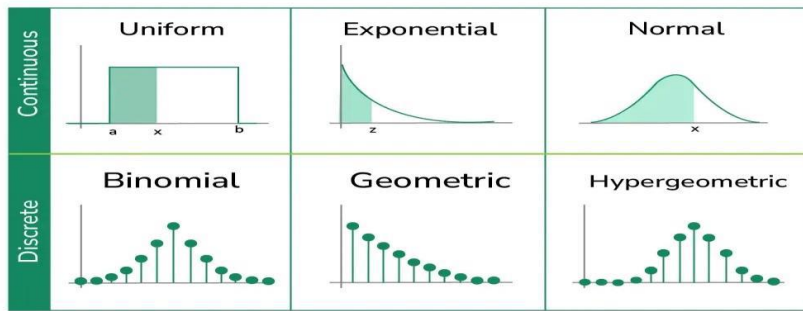
PROBABILITY DISTRIBUTION

Probability Distribution refers to the function $P(x)$ that distributes the probabilities of all possible values that the random variable 'X' can take. It depends on whether the random variable is discrete or continuous.

While a **frequency distribution** shows how often outcomes occur in a sample or dataset, a **probability distribution** assigns probabilities to outcomes. It help describe how data is spread, predict outcomes, identify rare events (outliers), and provide the foundation for statistical tests and modeling, such as regression analysis.

Discrete probability distribution is mutually exclusive i.e. the random variable takes distinct values (like the outcome of rolling a die). In a **continuous probability distribution**, the random variable can take any value within a certain range (like the height of a person).

Probability Distribution



1. Probability Distributions for Discrete Random Variables

*It is a mutually exclusive list of all the possible numerical outcomes along with the probability of occurrence of each outcome.

*It is described by its **Probability Mass Function (PMF)**, which assigns a probability to each possible value of the variable.

$$P(X=x) = p(x), \text{ where } \sum p(x)=1.$$

The key properties of a PMF are:

- Each probability is non-negative.
- The sum of all probabilities is equal to 1.

Types of discrete probability distributions include the **binomial distribution**, **Poisson distribution**, and **geometric distribution**.

Example:

Let X a random variable, represents no. of heads when three coins are tossed.

$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$

No. of heads(X)	0	1	2	3
Probability P(X)	1/8	3/8	3/8	1/8

Note: Each probability is >0 and their sum is 1.

A bar graph is appropriate with y axis measuring probability and x axis denoting the values.

2 Probability Distribution Function for continuous variable

A probability distribution function **for a continuous variable** describes the likelihood of the variable taking values within a certain range, using a mathematical function known as the probability density function (PDF).

* The probability at any point is 0. Area under curve only makes sense. The total probability (area under the curve for continuous) equals 1.

* For continuous variable, the probability of taking any exact single value $P(X=x)$ is always zero. Only intervals have non-zero probability.

* The cumulative probability is equal to 1 i.e. sample space.

* The probability density can never be negative, and can exceed 1.

* Area is a measure of probability

***Probability Density Function**

For a variable which can take any value in a given **interval**, the probability density function $f(x)$ is defined as

$$f(x) \geq 0, \int_{-\infty}^{\infty} f(x) dx = 1$$

(The total probability (area under the curve for continuous) equals 1)

(Integration is the process of summing or accumulating values over a continuous range.)

Probability for a range of values $[a, b]$ is: $P(a \leq x \leq b) = \int_a^b f(x) dx$

If x can take values in the interval $[a, b]$ then $\int_a^b f(x) dx = 1$

(For the sake of uniformity the range of every continuous random variable is taken to be $(-\infty, \infty)$, by taking the value of $f(x)$ to be 0 outside the actual range.)

For continuous variable, the probability of taking any exact single value $P(X=x)$ is always zero. Only intervals have non-zero probability.

Cumulative Distribution Function (CDF)

- The CDF, $F(x)$, gives the probability that X is less than or equal to x :

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

Range: The output values always fall between 0 and 1 ($0 \leq F(x) \leq 1$), as probability never exceeds 1

Limit at Negative infinity: $\lim_{x \rightarrow -\infty} F(x) = 0$ (Probability towards $-\infty = 0$)

Limit at Positive infinity: $\lim_{x \rightarrow \infty} F(x) = 1$ (Probability towards $+\infty = 1$)

CDF is monotonically increasing, as x increases CDF never decreases.

Example: Let a dice is rolled. $S = \{1, 2, 3, 4, 5, 6\}$

X	1	2	3	4	5	6
P(X≤ A)	P(x≤1) = 1/6	P(x≤2) = 2/6	P(x≤3) = 3/6	P(x≤4) = 4/6	P(x≤5) = 5/6	P(x≤6) = 6/6

A histogram is appropriate with y axis measuring probability.

Visualizing a PDF

1. **Discrete:** Bars indicating probabilities at specific points.
2. **Continuous:** Smooth curves showing the density of probabilities across the domain.

MEAN AND VARIANCE OF A PROBABILITY DISTRIBUTION

The **mean** and **variance** are the two most important measures that describe a probability distribution. The calculation methods depend on whether the random variable is **discrete** or **continuous**.

The mean μ , also known as **Expected Value E(X)**, is the average value of random variable when random experiment is repeated many times.

It is the weighted average of the variables where probability is the weight.

Distribution Type	Formula	Description
Discrete	$\mu = E(X) = \sum_x xP(X = x)$	The sum of each possible value (x) multiplied by its corresponding probability ($P(X = x)$).
Continuous	$\mu = E(X) = \int_{-\infty}^{\infty} xf(x)dx$	The integral of each possible value (x) multiplied by the Probability Density Function ($f(x)$) over all possible values.

The variance, σ^2 or $Var(X)$, is a measure of the **spread** or **dispersion** of the distribution. It tells you how far the values of the random variable are scattered around the mean (μ).  

Variance and standard deviation of a discrete random variable

$$Var(x) = \sigma^2 = \sum p(x)(x - \mu)^2 \text{ OR } \sum px^2 - \mu^2$$

$$S.D. = \sqrt{var(x)}$$

Variance and standard deviation of a continuous random variable

$$Var(x) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

Some properties of Expectation (mean) are as follows:

- In general, for any function $f(x)$, the expectation is $E[f(x)] = \sum f(x) \times P(X = x)$
- If k is a constant then $E(k) = k$
- If k is a constant and $f(x)$ is a function of x then $E[k f(x)] = k E[f(x)]$
- Let c_1 and c_2 be constants and u_1 and u_2 are functions of x then $E = c_1 E[u_1(x)] + c_2 E[u_2(x)]$

Properties of Variance

- As Variance is a statistical measure of the variability or spread of a set of data, thus the variance of a constant is 0 i.e., $\text{Var}(k) = 0$ when k is a constant.
- Variance is always non-negative because it is the expected value of a squared quantity. A variance of zero occurs only when all data points or outcomes are identical (no variability).
- If a random variable is changed linearly there is no effect of it on the variance i.e., $\text{Var}[X+a] = \text{Var}[X]$.
- If a random variable is multiplied by some constant then Variance is scaled by 2 i.e. $\text{Var}[aX] = a^2 \text{Var}(X)$.
- Combining the above two properties to get, $\text{Var}[aX + b] = a^2 \text{Var}(X)$

Example: Consider the following problem where a standard die is being rolled and it's been given that the probability of any face is proportional to the square of the number obtained on its face. Obtain the probability distribution table associated with this experiment

Solution:

Since $P(x) \propto x^2 \Rightarrow P(x) = kx^2$ where k is the proportionality constant and x are the numbers from 1 to 6

Also from the two conditions listed above we know that the sum of all probabilities is 1

Thus $\sum kx^2 = 1$ when summation is varied over x

$$\Rightarrow k + 4k + 9k + 16k + 25k + 36k = 1$$

$$\Rightarrow 91k = 1$$

$$\Rightarrow k = 1/91$$

Thus substituting this value of k into each probability we can obtain the probability of each random variable and then forming a table we get

X	1	2	3	4	5	6
P(x)	1/91	4/91	9/91	16/91	25/91	36/91

Example: What is the expectation when a standard unbiased die is rolled?

Solution:

Rolling a fair die has 6 possible outcomes : 1, 2, 3, 4, 5, 6 each with an equal probability of 1/6

Let X indicate the outcome of the experiment

$$\text{Thus } P(X=1) = 1/6$$

$$\Rightarrow P(X=2) = 1/6$$

$$\Rightarrow P(X=3) = 1/6$$

$$\Rightarrow P(X=4) = 1/6$$

$$\Rightarrow P(X=5) = 1/6$$

$$\Rightarrow P(X=6) = 1/6$$

$$\text{Thus, } E(X) = \sum x \times P(X=x)$$

$$\Rightarrow E(X) = 1 \times (1/6) + 2 \times (1/6) + 3 \times (1/6) + 4 \times (1/6) + 5 \times (1/6) + 6 \times (1/6)$$

$$\Rightarrow E(X) = 7/2 = 3.5$$

Example: Given $E(X) = 4$ and $E(X^2) = 6$ find out the value of $E(3X^2 - 4X + 2)$

Solution:

Using the various properties of expectation listed above, we get

$$E(3X^2 - 4X + 2) = 3 \times E(X^2) - 4 \times E(X) + E(2)$$

$$= 3 \times 6 - 4 \times 4 + 2$$

$$= 4$$

$$\text{Thus, } E(3X^2 - 4X + 2) = 4$$

Example : Find the variance and standard deviation of the following probability distribution table

X	0	1	2	3
P(X)	0.1	0.2	0.4	0.3

Solution:

First off, we need to check if this distribution table is valid or not.

So each $P(x) > 0$, and sum of all probabilities is 1

Thus, this is a valid distribution table.

Now to find variance, we need $E(X)$ and $E(X^2)$

$$E(X) = \sum x p(x) = 0 \times 0.1 + 1 \times 0.2 + 2 \times 0.4 + 3 \times 0.3$$

$$\text{Thus, } E(X) = 1.9$$

$$E(X^2) = \sum x^2 p(x) = 0 \times 0.1 + 1 \times 0.2 + 4 \times 0.4 + 9 \times 0.3$$

$$\text{Thus, } E(X^2) = 4.5$$

$$\text{Thus } \text{Var}(X) = 4.5 - (1.9)^2 = 0.89$$

$$\text{Standard deviation} = (0.89)^{0.5} = 0.9433$$

Example:

In a manufacturing unit inspection from a lot of 20 baskets which includes 6 defectives, a sample of 2 baskets is drawn at random without replacement. Prepare the probability distribution of the no of defective baskets. Also calculate mean and variance for random variable X.

Example:

Four bad oranges are accidentally mixed with 16 good oranges. Find the probability distribution of the no. of bad oranges in a draw of two oranges. Also find the mean, variance, standard deviation of the distribution.

TYPES OF DISCRETE PROBABILITY DISTRIBUTION

Bernoulli's Trials : Trials of random experiment are called bernoulli's trials if they satisfy the following conditions.

- a) They are finite in number.
- b) They are independent of each other
- c) Each trail has exactly two outcomes: success or failure
- d) The probability of success or failure remains same in each trial.

Binomial Distribution

A **Binomial Distribution** for a random variable $X = 0, 1, 2, \dots, n$ is defined as the probability of **success or failure** in a series of independent trials. Each trial is independent of the others, and the distribution helps calculate the probability of various outcomes in these trials.

(The probability of success or failure remains same in each trial)

The Binomial Distribution Formula which is used to calculate the probability, for a random variable $X = 0, 1, 2, 3, \dots, n$ is given as

$$P(X = r) = {}^nC_r p^r q^{n-r}, r = 0, 1, 2, 3, \dots$$

Where,

- **n** = Total number of trials
- **r** = Number of successes
- **p** = Probability of success
- **q** = Probability of failure ($q = 1 - p$)

So we can construct the following table:

$P(x = 0)$	$P(x = 1)$	$P(x = 2)$		$P(x = n-1)$	$P(x = n)$
q^n	${}^nC_1 p q^{n-1}$	${}^nC_2 p^2 q^{n-2}$		${}^nC_{n-1} p^{n-1} q$	p^n

```
</> Python
from scipy.stats import binom

n = 10
p = 0.5

prob = binom.pmf(6, n, p)
print(prob)
```

Example:

If a die is rolled 5 times, what is the probability

- a) of having 1 in two rolls.
- b) Having at most 2 ones
- c) Having at least 2 ones

Let p be the probability of getting 1, when a dice is rolled
 Here $n = 5$, $p = 1/6$, $q = 5/6$

- a) $P(X=2) = {}^5C_2 (1/6)^2 (5/6)^4 = 1250/7776 \approx 0.1608$
- b) $P(X \leq 2) = P(0) + P(1) + P(2)$

$$\frac{3125}{7776} + \frac{3125}{7776} + \frac{1250}{7776}$$

$$0.4019 + 0.4019 + 0.1608 = 0.9646$$
- c) $P(X \geq 2) = P(2) + P(3) + P(4) + P(5) = 1 - P(X < 2)$

$$= 1 - [P(0) + P(1)]$$

$$= 1 - (0.4019 + 0.4019)$$

$$= 1 - 0.8038 = 0.1962$$

Example:

Out of 800 families with 4 children each, what percentage would be expected to have:

- a) 2 boys 2 girls
- b) At least one boy
- c) No girls
- d) Atmost 2 girls

Example:

A factory produces light bulbs. Historically, 5% of the light bulbs are defective. If a quality control inspector randomly selects 10 light bulbs, what is the probability that exactly 2 of them are defective?

Example:

An online store finds that 20% of users who visit the website make a purchase. If 15 users visit the site, what is the probability that at least 3 users make a purchase?

Example:

On the basis of past experience Martin Cloth Store manager estimates that the probability of purchase by any customer is 0.3 and purchase decision of one customer does not influence the other. What is the probability of 2 of the next 3 customers will make a purchase?

Mean and Variance of Binomial Distribution

The mean of binomial distribution is np

$$\mu = \sum r P(r) = \sum r {}^nC_r p^r q^{n-r} = np$$

Variance of binomial distribution is $\sigma^2 = npq$

Standard Deviation = $\sqrt{npq}$

Example:

A digital marketing agency finds that 30% of users who click on their online advertisement make a purchase. If the advertisement is clicked by 20 users, calculate:

1. The **mean** number of users who make a purchase.
2. The **variance** of the number of users who make a purchase.

2.Poisson Distribution

Used to model **number of events occurring in a fixed interval of time or space.**

The Poisson Distribution is the limiting form of binomial distribution when number of trials, n is very large ($n \rightarrow \infty$) and probability of success is small ($p \rightarrow 0$) such that np always remains finite, say λ . So if $n \geq 50$ and $\lambda = np \leq 5$, the Poisson distribution gives good approximation of Binomial distribution.

The distribution is used to describe the behavior of rare events such as number of printing mistakes in a book, number of accidents on road.

The Poisson distribution is a probability distribution that helps us model the number of times an event happens in a fixed interval of time, space, or any unit (like area, volume, etc.), under certain conditions.

*The probability of occurrence is same in two intervals of same length.

*The probability of occurrence or non-occurrence is independent of occurrence or non-occurrence in any other interval.

BUSINESS EXAMPLES

- To count number of calls received at a call center per minute.
- To count number of patients arriving at an emergency room in an hour.
- * To count number of cars passing through a toll booth per hour.
- * To count the no. of particles emitted from radioactive substance.

Recurrence Formula for Probabilities of Poisson Distribution is:

$$P(r) = \frac{e^{-\lambda} \lambda^r}{r!} \quad \text{and} \quad P(r+1) = \frac{\lambda}{r+1} P(r)$$

where :

$P(r)$ = probability of r occurrences in an interval

r = event

Lambda (λ) = mean no. of occurrences in an interval

$e = 2.71828$

```

</> Python

from scipy.stats import poisson

lam = 3
prob = poisson.pmf(2, lam)

print(prob)

```

Mean of Poisson Distribution is λ

Variance of Poisson Distribution is λ .

Standard Deviation of Poisson Distribution is $\sqrt{\lambda}$

Example:

Imagine a bakery receives an average of 5 customers every hour. What's the probability that exactly 3 customers will arrive in the next hour?

Here:

$\lambda=5$ (average customers per hour).

$k= 3$ (we're looking for exactly 3 customers).

Using the formula:

$$P(X=3) = (5^3 e^{-5})/3! = 125 \cdot 0.0067 / 6 \approx 0.1404$$

So, the probability of exactly 3 customers arriving is approximately **14.04%**.

Example:

If the probability that an individual suffers a bad reaction from injection of a given serum is 0.001, determine the probability that out of 2000 individuals

- a) Exactly 3 b) more than 2 individuals suffer from bad reaction. (use $e^{-2} = 0.1353$)

Example:

A traffic engineer records the number of bicycle riders that use a particular cycle track. He records that an average of 3.2 bicycle riders use the cycle track every hour. Given that no. of bicycles that use the cycle track follow a Poisson distribution, what is the probability that:

- a) 2 or less bicycle riders will use the cycle track within an hour
b) 3 or more bicycle riders will use the cycle track within an hour

Also write the mean expectation and variance of random variable X.

Example:

Twenty sheets of aluminum alloy were examined for surface flaws. The frequency of the number of sheets with a given number of flaws per sheet was as follows:

No. of flaws	0	1	2	3	4	5	6
Frequency	4	3	5	2	4	1	1

What is the probability of finding a sheet chosen at random which contains 3 or more surface flaws?

Example:

Vehicles pass through a junction on a busy road at an average rate of 300 per hour.

- Find the probability that none passes in a given minute.
- What is the expected number passing in two minutes?
- Find the probability that this expected number actually pass through in a given two-minute period.

3.Geometric Distribution

Geometric distribution models the number of trials needed to get the **first success** in repeated Bernoulli trials (yes/no, success/failure), where each trial is independent and has the **same** probability of success p .

Examples:

- Number of coin tosses needed to get the first head
- Number of attempts before a basketball player scores
- Number of customers contacted before making the first sale

It counts trials until the first success, models “how many attempts needed?” It is **memory less** like exponential distribution (**Past failures do not affect future probabilities.**)

Example:

You toss a biased coin where probability of heads = 0.3.

Let X = trial number on which you see the **first head**.

Then X follows a geometric distribution with parameter $p = 0.3$.

Formula to find probability to get success in k th trial is:

$$P(X = k) = (1 - p)^{k-1} p$$

Interpretation:

- $P(X = 1) = p$: success on the first trial.
- $P(X = 2) = (1 - p)p$: fail first, succeed second.
- $P(X = 3) = (1 - p)^2 p$: fail, fail, then succeed.

$$\text{Mean} = 1/p, \text{ variance} = (1 - p)/p^2$$

Ques: Suppose a coin has probability of getting heads as 0.5, What is the probability of getting the first head on the 3rd toss?

As $p = 0.5$, using the formula we get,

$$\begin{aligned}
 P(X = 3) &= (1 - 0.5)^{3-1}(0.5) \\
 &= (0.5)^2(0.5) \\
 &= 0.125 \quad \text{so probability is 0.125 or 12.5\%}
 \end{aligned}$$

Use **geometric distribution** when:

- Each trial has **two outcomes**: success or failure.
- Probability of success on each trial is **constant** = p .
- Trials are **independent**.
- The random variable X = number of trials **up to and including the first success**.

Python

```

from scipy.stats import geom

p = 0.4
prob = geom.pmf(3, p)

print(prob)

```

TYPES OF CONTINUOUS PROBABILITY DISTRIBUTION

A continuous probability distribution describes probabilities for a **continuous random variable** using **Probability Density Function (PDF)** where:

- $f(x) \geq 0$
- Total area under curve = $\int f(x)dx = 1$
- $P(a \leq x \leq b) = \int_a^b f(x)dx$

Important distributions are:

1. **UNIFORM DISTRIBUTION**

The uniform distribution is a continuous probability distribution where every outcome within a specified range is equally likely. Think of it like picking a random point on a straight line segment—every point has the same chance of being selected.

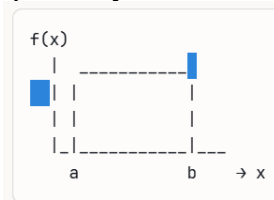
Example:

Selecting a random number between 0 and 1

The arrival time of a bus that comes every 15 minutes—if you arrive at a random time, your wait time is uniformly distributed between 0 and 15 minutes

- Every value in the interval $[a, b]$ is equally likely.
- PDF = $f(x) = 1/(b - a)$, for $a \leq x \leq b$, where a is min value and b is max value.
- Mean = $\mu = (a + b) / 2$
- Variance = $\sigma^2 = (b-a)^2 / 12$

- Shape : perfectly flat or rectangular



Python

```
from scipy.stats import uniform

data = uniform.rvs(size=1000, loc=0, scale=10)
```

Where:

rvs:	generates random samples
Size (1000):	is the number of random values generated
Loc(0):	is the starting value of distribution (minimum)
Scale (10):	is the range width

Example: A bus arrives every 10 minutes. The time a person waits for the bus (randomly arriving) is uniformly distributed between 0 and 10 minutes. ($p = 0$, $Q=10$)

Probability: If waiting time is between 2 and 5 minutes: ($b-a$ is $10-0$)

$$P(2 \leq X \leq 5) = \int_2^5 \frac{1}{10} dx = \frac{3}{10} = 0.3$$

EXPONENTIAL DISTRIBUTION

The exponential distribution models the time between events that occur continuously and independently at a constant average rate. It's commonly used when events follow a Poisson process.

Ex: time between customer arrivals, time between phone calls..

- Probability density function: $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$, where λ is the rate parameter (average events per unit time)
- Mean: $\frac{1}{\lambda}$ (average time between events)
- Variance: $\frac{1}{\lambda^2}$
- **Memoryless property:** The probability of an event occurring in the next interval is independent of how much time has already passed

Python

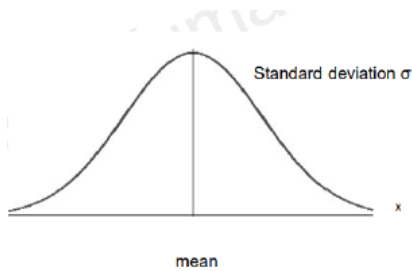
```
from scipy.stats import expon

data = expon.rvs(scale=2, size=1000)
```

Uniform distribution assigns equal probability across a range, while exponential distribution models waiting times where shorter intervals are more likely than longer ones.

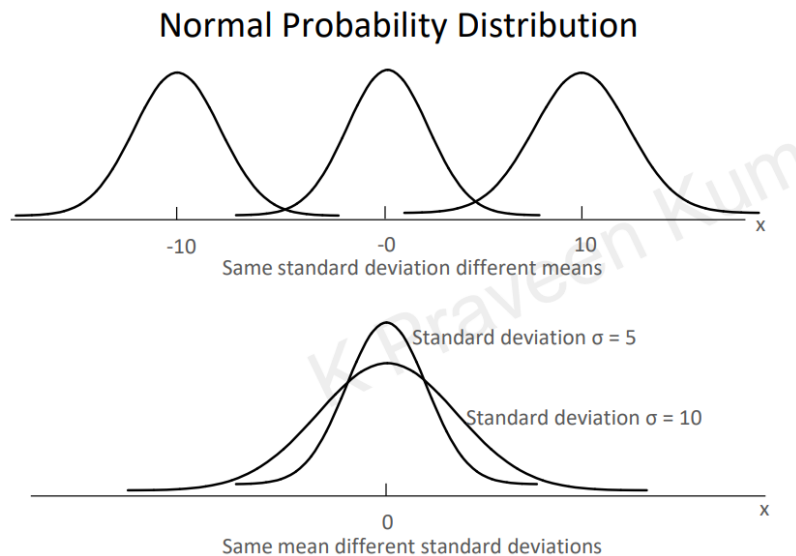
NORMAL DISTRIBUTION

Normal Distribution is the most important continuous probability distribution that describes the data that cluster around mean (μ) with a symmetrical bell-shaped curve. i.e. **it is a bell shaped curve symmetrical about a vertical line through the mean.** It is often referred to as **Gaussian Distribution**. It is an approximation to Binomial Distribution.



* The entire family of normal probability distribution is differentiated by its **mean** and **standard deviation**.

*The total area under the normal curve is 1.

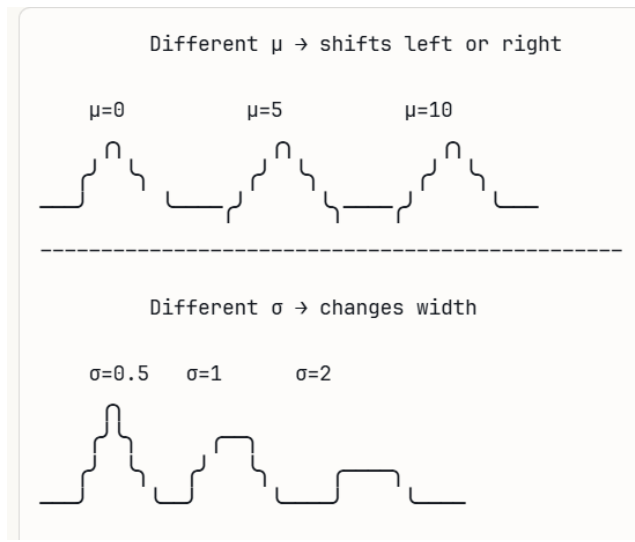


The density function of a normal distribution is called Normal Density function given by

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad \text{for } -\infty < x < \infty$$

where

- μ is mean (*centers the distribution*)
- σ is standard variation (*controls width/spread*)
- $e = 2.7183$
- $\sqrt{2\pi} = 2.5066$



Python

```
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

data = np.random.normal(50, 10, 1000)

sns.histplot(data, kde=True)
plt.show()
```

Features of Normal Distribution

1. Shape:

- The graph is a symmetric bell- shaped curve.
- The highest point is at the mean (μ), which is also the median and mode of the distribution

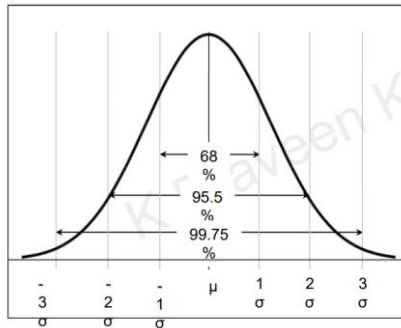
2. Mean, Median, Mode:

- All are same and lie at the center of distribution.
mean = median = mode

3. Spread:

The normal distribution is differentiated by its mean and standard deviation.
The spread is determined by the **standard deviation** (σ).

- About 68% of the data falls within 1 standard deviation ($\mu \pm \sigma$).
- About 95% of the data falls within 2 standard deviations ($\mu \pm 2\sigma$).
- About 99.7% of the data falls within 3 standard deviations ($\mu \pm 3\sigma$).



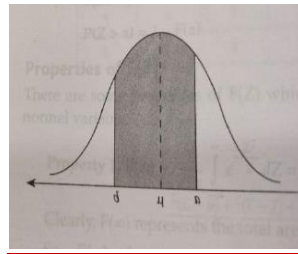
4. Symmetry:

The normal curve is symmetric on y- axis, bisected by mean. The left and right sides of the curve are mirror images.

The tails of curve never crosses the x-axis and extends indefinitely on right hand and left hand side.

5. Probabilities:

- Probabilities are calculated as the area under the curve for a given range of values.
- Area under normal curve bounded by ordinates $x = a$, $x = b$ is equal to the probability of random variable X between $x = a$, $x = b$



Examples of normal Distribution: (each normal distribution is with its own mean and spread)

- IQ scores ($\mu = 100$, $\sigma = 15$)
- Adult male height ($\mu = 175$, $\sigma = 7$)
- Exam scores ($\mu = 70$, $\sigma = 10$)

TESTS TO CHECK NORMALITY OF DATA

Two of the most important normality tests available in SciPy are:

1. Shapiro–Wilk Test

The Shapiro–Wilk test is a normality test. It checks whether your sample data could reasonably have come from a **Normal Distribution** (bell curve).

It compares the **actual ordered sample values** with the **expected values from a normal distribution**. (check similarity between two)

```
# Perform Shapiro test
stat, p_value = stats.shapiro(data)
```

The Shapiro–Wilk test computes a statistic W (between 0 and 1) and p_value (to accept or reject normality).

W close to 1: data is normal

p-value > 0.05: data is normal

W far from 1: data may not be normal.

P-value ≤ 0.05: data is not normal

- Very effective for **small to moderate sample sizes** (say n from about 3–50);
- One of the most powerful normality tests
- Commonly used in statistical analysis and machine learning

2. D’Agostino–Pearson Normality Test

The D’Agostino–Pearson test is also a normality test, but unlike Shapiro–Wilk (which looks at the whole shape in a single statistic), this one explicitly analyze **skewness** (asymmetry) and **kurtosis** (tail/peak shape).

A normal distribution should have skewness near 0 and kurtosis near 3

The test combines both measurements into one test statistic K^2 .

```
# Perform D’Agostino-Pearson Test
stat, p = stats.normaltest(data)
```

K^2 (stat) measures how far skewness and kurtosis are from their normal values.

stat (small value): data is close to normal

p-value > 0.05: data is normal

stat (large value): Data is far from normal

P-value ≤ 0.05: data is not normal

large stat and small p-value: data not normal; small stat and large p-value: data is normal

Important Points

- Better for large datasets
- Detects asymmetry and unusual tails
- Widely used in data science projects

Simple Difference

Feature	Shapiro-Wilk	D'Agostino-Pearson
Best for	Small datasets	Large datasets
Based on	Ordered sample values	Skewness & kurtosis
Accuracy	Very high	Good for large samples
SciPy Function	shapiro()	normaltest()

These tests are fundamental in Statistics, Machine Learning, and predictive analytics before applying many machine learning algorithms.

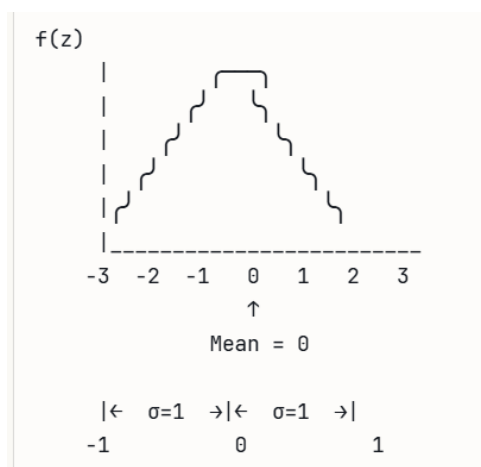
Standard Normal Distribution

A random variable that has a normal distribution **with mean $\mu=0$ and standard deviation $\sigma=1$** is said to have standard normal distribution and is represented by **z**.

The standard normal distribution is symmetric and bell shaped centered at 0. Total area under its curve is 1, representing the total probability.

So, the density function of standard normal variable **z** is given by:

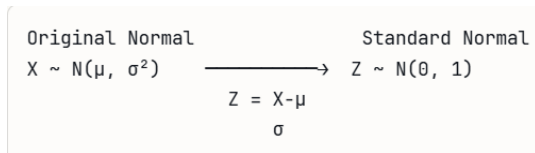
$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \quad , -\infty < z < \infty$$



Note: To find **area under any normal curve** first convert the normal distribution random variable **X** to standard normal distribution variable **Z** by the formula

$$Z = \frac{X - \mu}{\sigma} \quad \text{where } x \text{ is the original value, } \mu \text{ is mean, } \sigma \text{ is standard deviation.}$$

It provides a standard way to understand and calculate probabilities for normal distribution by converting values into standardized form called **z-scores**.



A z-score tells you how many standard deviations a value is from the mean. (**$z=0$** means exactly at mean; **$z=1$** is 1 σ above mean; **$z = -1$** means 1 σ below mean)

The z score is looked up in z table.

Note: For normal distribution, different table are referred for different μ and σ but for standard distribution only z table is referred. (**Standard normal distribution acts as a universal reference for ALL normal distributions**)

Features of the Standard Normal Distribution

1. **Shape:**

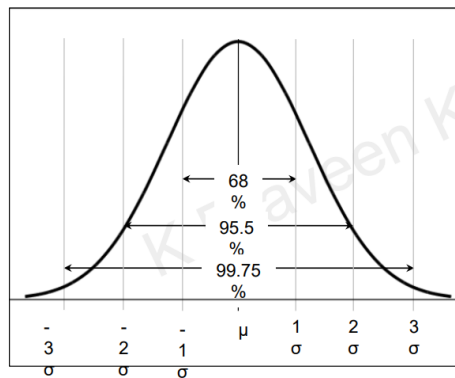
- The graph is a symmetric, bell-shaped curve.
- The highest point is at the mean (μ).

2. **Mean, Median, and Mode:**

- These are all the same and lie at the center of the distribution.

3. **Spread:**

- The spread is determined by the **standard deviation (σ)**.
- About 68% of the data falls within 1 standard deviation ($\mu \pm \sigma$).
- About 95% of the data falls within 2 standard deviations ($\mu \pm 2\sigma$).
- About 99.7% of the data falls within 3 standard deviations ($\mu \pm 3\sigma$).
- This is known as 68 – 95 – 99.7 rule or Empirical Formula.

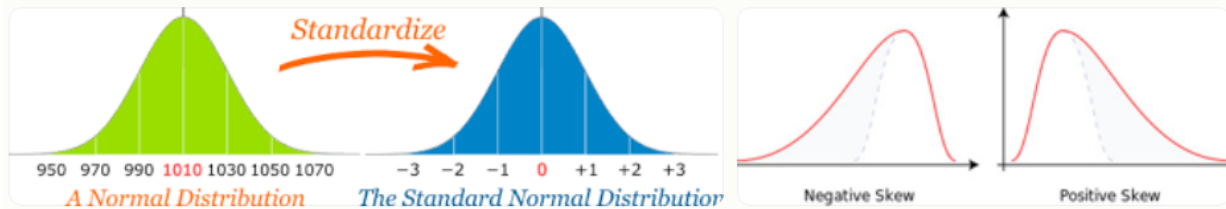


The total area under curve is always summed to 1.

4. **Probabilities:**

- Probabilities are calculated as the area under the curve for a given range of values.
- Area under normal curve bounded by ordinates $x = a$, $x = b$ is equal to the probability of random variable X between $x = a$, $x = b$.

difference between normal distribution and standard normal distribution



The difference between a **normal distribution** and a **standard normal distribution** lies primarily in the values of their parameters—mean and standard deviation:

Aspect	Normal Distribution	Standard Normal Distribution
Mean (μ)	Can be any real number (e.g., 10, 50).	Always 0.
Standard Deviation (σ)	Can be any positive value (e.g., 2, 5).	Always 1.
Definition	A continuous probability distribution characterized by its mean and standard deviation; has a bell-shaped curve.	A special case of the normal distribution with mean 0 and standard deviation 1; called the Z-distribution.
Use	Models real-world data with varying centers and spreads.	Provides a standard reference scale for all normal distributions; used to calculate probabilities through z-scores.
Transformation	Any normal distribution can be converted to a standard normal distribution using the formula:	

CENTRAL LIMIT THEOREM

It is a fundamental concept that explains why the normal distribution appears so frequently in real world analysis.

It states that if you take many random samples from any population regardless of population's distribution shape, and calculate means of these samples then, as the sample size n increases, the distribution of these sample means approaches a normal distribution about the population mean, even if the population itself is not normally distributed. i.e. 95% of all sample means should fall within 2σ of the population mean.

The standard deviation equals to the population standard deviation divided by the square root of the sample size $\sigma / \sqrt{n}$

Why is it important?

- The CLT allows us to use normal probability models for inference about population means even when the population distribution is not normal.
- It forms the theoretical basis for many statistical methods, including confidence intervals and hypothesis testing.
- It explains why averages of measurements tend to be normally distributed, even if the original data are skewed or irregular.

Mathematical Summary:

If $X_1, X_2, \dots, X_n$ are independent, identically distributed random variables with mean μ and finite variance σ^2 , then the sample mean $\bar{X}$ tends toward a normal distribution as n becomes large:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$